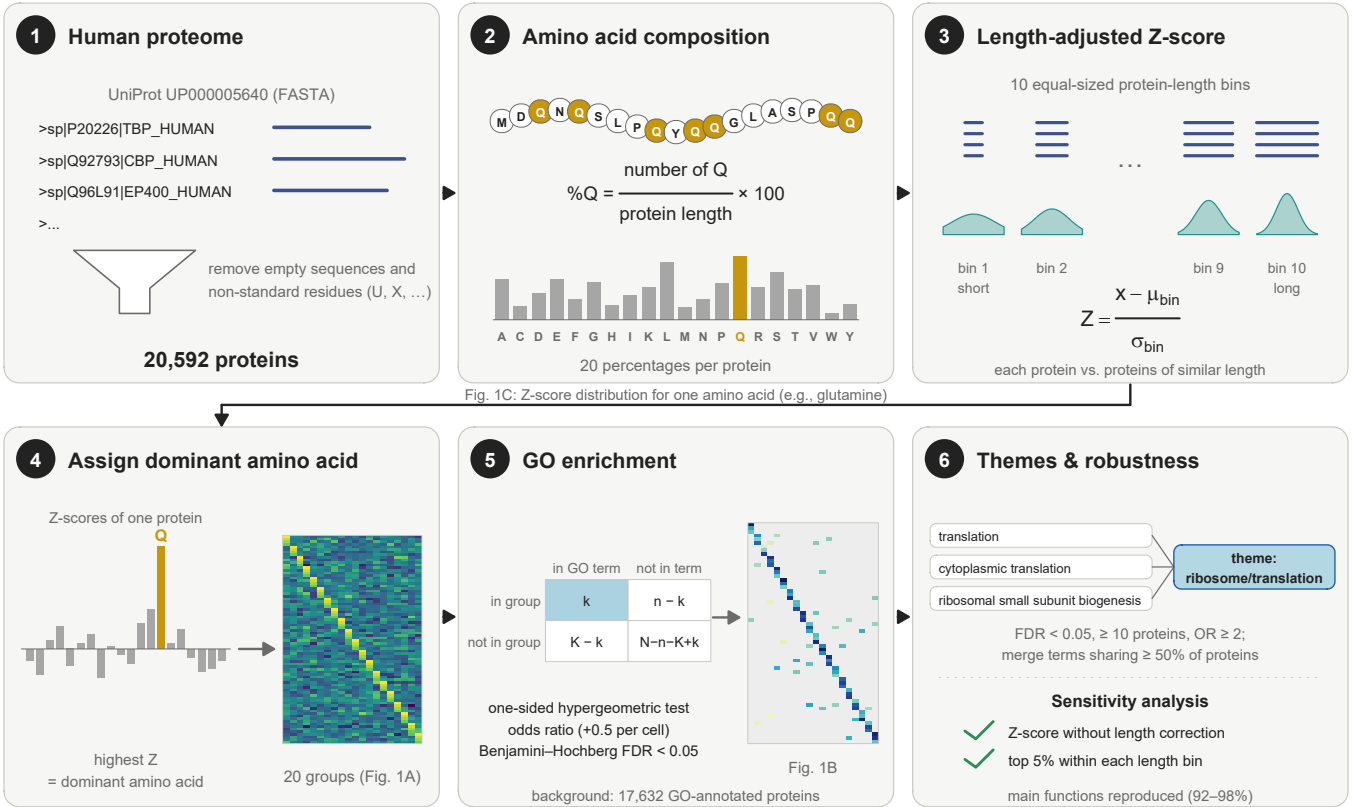

1
Human proteome
UniProt UP000005640 (FASTA)
>sp|P20226|TBP_HUMAN
>sp|Q92793|CBP_HUMAN
>sp|Q96L91|EP400_HUMAN
>...
remove empty sequences and non-standard residues (U, X, …)
20,592 proteins
2
Amino acid composition
number of Q
%Q = ——————— × 100
protein length
A C D E F G H I K L M N P Q R S T V W Y
20 percentages per protein
3
Length-adjusted Z-score
10 equal-sized protein-length bins
bin 1
short
bin 2
bin 9
bin 10
long
Z = (x − μbin) / σbin
each protein vs. proteins of similar length
Fig. 1C: Z-score distribution for one amino acid (e.g., glutamine)
4
Assign dominant amino acid
Z-scores of one protein
Q
highest Z
= dominant amino acid
20 groups (Fig. 1A)
5
GO enrichment
in GO term
not in term
in group
k
n − k
not in group
K − k
N−n−K+k
one-sided hypergeometric test
odds ratio (+0.5 per cell)
Benjamini–Hochberg FDR < 0.05
Fig. 1B
background: 17,632 GO-annotated proteins
6
Themes & robustness
translation
cytoplasmic translation
ribosomal small subunit biogenesis
theme:
ribosome/translation
FDR < 0.05, ≥ 10 proteins, OR ≥ 2;
merge terms sharing ≥ 50% of proteins
Sensitivity analysis
Z-score without length correction
top 5% within each length bin
main functions reproduced (92–98%)